3 A comparative study of wellbeing in students

Tecmilenio case

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Introduction

This study focuses on the impact of the health emergency due to coronavirus disease 19 (COVID-19), which is caused by the SARS-CoV-2 virus, on the students at Tecmilenio University. The following are some questions of interest addressed in this research: What is the impact of COVID-19 on the wellbeing of students at Tecmilenio University? What is the effect of COVID-19 on the positive emotions of students? Is there a difference between academic levels? This topic is relevant because it allows identifying whether being immersed in an ecosystem of wellbeing fosters greater resilience skills in students to help them better carry the emotional burden posed by a health emergency. It is expected that the present study will identify a moderate impact of the COVID-19 health emergency on student wellbeing. The methodology used for this work included a wellbeing survey conducted during the month of April in 2018, 2019 and 2020. The survey is representative of the student population, which is approximately 60 thousand students.

The results indicate that the Wellbeing and Happiness Ecosystem at Tecmilenio University (*Universidad Tecmilenio*) is fulfilling its mission of providing students with tools to face various personal and professional challenges. The students reported fewer negative emotions in 2020 than in 2019. The finding merits greater importance when considering that other surveys conducted in Mexico by other universities indicated that the country had experienced an increase in emotions such as anxiety and fear.

This chapter is divided into four sections: the first section includes a theoretical review, empirical background and a description of the Wellbeing and Happiness Ecosystem at Tecmilenio University; the second section describes the data and the methodology used; the third section presents the results of the study; and the final section presents the conclusions.

Objectives

The objectives of the study are to identify the impact of COVID-19 on the wellbeing of students at Tecmilenio University and explore the differences between high school and college students.

Theoretical framework

As of August 2020, Tecmilenio University has more than 60,000 students, more than 100,000 graduates and more than 5,800 professors. Approximately 30% of the enrolled students are high school students, 30% are professional-level students taking in-person classes, 20% are professional-level students taking online classes and 20% are master's students. The university currently has 30 in-person campuses throughout 25 cities in Mexico and is about to open the 31st in-person campus. In addition, the online campus serves almost one-fifth of current students in undergraduate programmes and master's programmes, in particular, the master's degree in management.

The university has a nonprofit status and is part of an educational group led by the Monterrey Institute of Technology and Higher Education (*Instituto Tecnológico y de Estudios Superiores de Monterrey*), an institution with 77 years of leadership in education in Mexico. This year marks the 18th year, the university has operated with the interest of expanding educational coverage to populations that usually only have access to public universities. Thirty-three percent of university students are the first generation in their family to pursue a bachelor's degree and 70% have some type of financial support or scholarship.

Children and adolescents require emotional skills training, and there are different conceptual frameworks on how to teach these so-called soft skills or life skills. The Organisation for Economic Co-operation and Development (OECD) presents a conceptual framework on what children should learn; the framework includes wellbeing because it is considered a personal mastery skill that leads to significant learning opportunities. According to Peterson (2006), universities should provide students with much more than professional skills and abilities. Universities must have ethical goals that guide their students to be compassionate and seek intellectual, social and emotional growth. Most of the universities do not have wellbeing programmes for their students (Waters et al., 2012). In 2013, Tecmilenio University in Mexico generated a wellbeing model, based on the sense of life purpose, as a mandatory part of the university-level curriculum. The English economist Richard Layard argues that each of us, in all our decisions, must generate the greatest possible happiness (Layard, 2020). In Layard's view, today we have enough information to make evident the need to work with schools on these issues. A large number of people, millions, who engage in meditation services or holistic practices worldwide, the growth in coaching services and social support groups, labour policies promoting stress management, are evidence of the underlying need to address these issues. Happiness can be seen as an indicator of the progress of societies (Charles-Leija et al., 2018; Rojas & Charles-Leija, 2021).

If we further consider the high dropout rate and the increase in depression and anxiety diagnoses among the young population, dealing with issues of emotional wellbeing is a prevailing need not only for basic education schools but also for universities (Oades et al., 2011). In Mexico, the dropout rates are 15.5% for high school and 38% for universities.

Positive psychology presents one option for training in emotional wellbeing. In 2019, the most popular class at Yale University was "Happiness and a good life" but that is not the only evidence of need that exists. An important question in education is what kind of people do we want to shape? One need that became evident in 2020 is the need to manage emotions, understand and care for others and understand and protect mental health. Positive interventions should be applied not only at the curriculum level but also in a transversal manner and should change educational institutions (Peterson, 2006). More than creating isolated courses, such as at Yale, the challenge has been for Tecmilenio University to have a comprehensive strategy for the entire educational community, that is, students, professors and administrators, even extending further to parents and graduates.

Tecmilenio University, with almost 60 thousand students, integrates the principles of wellbeing and happiness in its educational philosophy as an integral part of its model and culture. The new university model includes a curriculum created to incorporate this comprehensive view of wellbeing into the baccalaureate, bachelor and graduate degree levels. Additionally, student life programmes aim to generate life skills, and a training programme has been created for professors and collaborators. A true transformation of the educational model and the institution occurred based on wellbeing. The 2020 vision established by the university was to train people with life purpose and the skills to achieve it.

A key planning approach for the year 2020 was the idea that human beings are living longer, up to 35 years longer than the generation of our grandparents, and that university education can contribute to creating skills that go far beyond work or professional life. To design the model, a working group consulted with experts who had shown evidence of the benefits of wellbeing and happiness and their positive correlation with a long, successful and satisfactory life. Additionally, it has been shown in several studies in Bhutan, Peru and Mexico that there are immediate benefits in the school performance of children and young people who receive training in emotional wellbeing and resilience (Seligman & Adler, 2019).

In 2013, the university created the Institute for Wellbeing and Happiness, signifying that wellbeing was a central project of the university. The Institute was born with four objectives: (a) teach wellbeing through courses and offer certificates and postgraduate degrees in positive psychology; (b) exemplify wellbeing in our daily life at the university by applying what we know about wellbeing; (c) provide services for wellbeing through consultations, training and coaching for organizations and companies based on positive psychology; and (d) investigate wellbeing by applying knowledge about happiness and how to enhance it.

From 2013 to date, a longitudinal study has been carried out on university students seeking to measure the shifts in wellbeing based on the PERMA (Positive Emotion (P), Engagement (E), Relationships (R), Meaning (M), Accomplishment (A)) model and seeking to gather evidence of the positive effect of the creation and implementation of the Wellbeing and Happiness Ecosystem.

To carry out this approach towards wellbeing and happiness within the university, the Wellbeing and Happiness Ecosystem was created on the basis of the elements of the PERMA model (Seligman, 2012) and components of physical wellbeing and mindfulness (Goleman & Davidson, 2017); additionally, the model was framed around developing character strengths (Peterson & Seligman, 2004). The ecosystem is represented graphically in Figure 3.1; the core aspects of the university are placed in the centre (Figure 3.1).

The multidimensional model emphasizes the main pathways of inserting wellbeing into the institution as a whole: (a) introducing wellbeing and happiness as a compulsory subject at the high school and professional levels and incorporating it into the business semester (professional practices) of undergraduate students; (b) reconfiguring student activities as wellbeing activities (including sports, culture and student clubs) and generating a sense of collaboration with the community through community service; (c) ensuring the application of the components of wellbeing through the training of all professors and creating minimum standards; and finally, (d) making changes in the facilities, such as redesigning cafeterias, gyms and libraries, to generate a culture of wellbeing.

As part of the curriculum at the high school and professional levels, all first-year university students are exposed to wellbeing and happiness content. At the

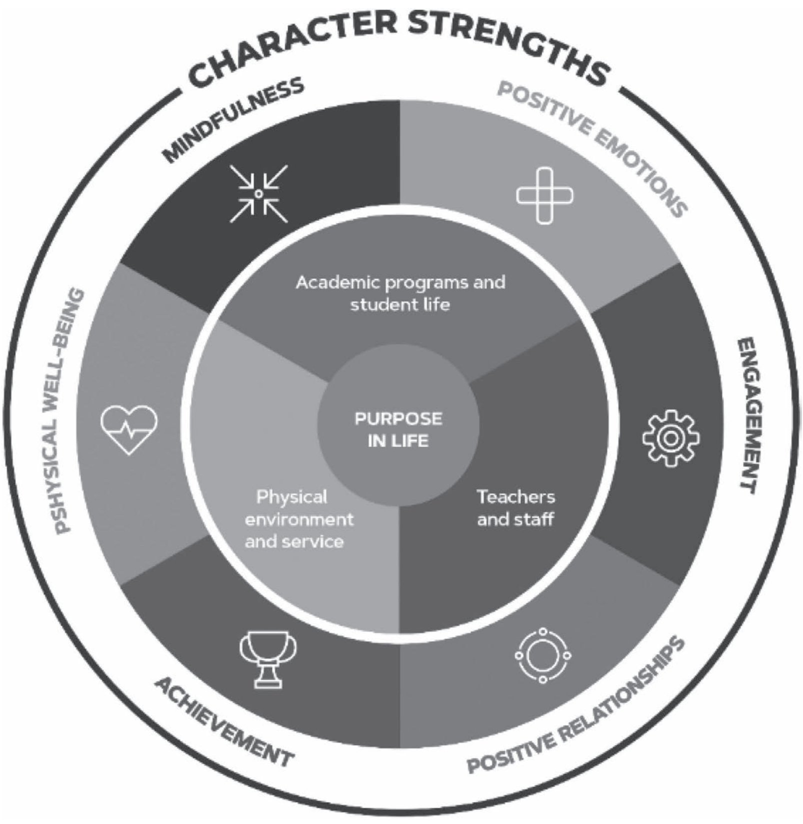

Figure 3.1 Wellbeing and happiness ecosystem

high school level, the content is distributed through six courses called Skills and Values. At the undergraduate level, there are four important curricular moments. Introduction to the University and Principles of Wellbeing and Happiness are studied in the first semester of the first year. For the proper functioning of the model in the ecosystem, it is essential that students are aware of and are owners of their own wellbeing, know how to identify the elements that compose wellbeing and have their own tools to utilize in specific contexts. They are even invited to teach others these tools on two occasions: the first occurs through community service (mandatory element of all degree programmes in Mexico) and the second occurs during the business semester or through internships, where students generate a positive intervention in their workplace.

The age of the student population was a determining factor in establishing a differentiated curriculum between high school and professional-level students. High school students receive content distributed in six subjects throughout high school according to age (Table 3.1):

The model for undergraduate has four stages that will be described in this section. The first one is just learning theory, the last one is teaching others.

Table 3.1 Summary description of skills and values contents

Course (Term)		*Wellbeing Dimension*	*Description and Source*
I	Skills and Values I (First Term)	Introduction to the Wellbeing and Happiness' Ecosystem in Tecmilenio	Students know the Wellbeing and Happiness' Ecosystem and learn on its different dimensions.
		Active constructive responding	Students practise how to react in a visibly positive and enthusiastic way to good news from someone else (Gable et al., 2004).
		Physical wellbeing and healthy habits	From a theoretical perspective, students learn on the benefits of promoting physical wellbeing and developing healthy habits (i.e. healthy food, exercise & sleep).
		Personal strengths	Students identify their personal strengths by completing the VIA-IS scale (Peterson & Seligman, 2004).
II	Skills and Values II (Second Term)	Personal strengths	Parents also complete the Values in Action scale so students may identify family strengths of character.
III	Skills and Values IV (Fourth Term)	Meaning	Students complete some exercises to start defining their purpose in life (Steger et al., 2009).

(*Continued*)

Table 3.1 (Continued)

Course (Term)		*Wellbeing Dimension*	*Description and Source*
		Goals	Students set self-concordant goals (setting goals according to their purpose in life) (Sheldon, 2014). They also practise a SOAR analysis based on appreciative inquiry (Stravos et al., 2003).
		Gratitude	Students learn on the benefits of gratitude and practise the three blessings exercise (Seligman, 2012).
V	Skills and Values V (Fifth Term)	Positive emotions	Students define and identify positive emotions (Seligman, 2012).
		Flow	Students learn what flow is and identify moments in their life when they experience (Wesson & Boniwell, 2007).
VI	Skills and Values VI (Sixth Term)	Goals	Students practise again SOAR analysis (Stravos et al., 2003). They learn how to set goals with mental contrasting techniques (Duckworth et al., 2011).
		Gratitude	Students express gratitude to others (e.g. parents or teachers).

Source: VIA-IS: Values in Action Inventory of Strengths.

(a) The introduction to the University course

The Introduction to the University course serves as a guide to contact a mentor or counsellor, who guides students in learning everything that the university has to offer within its flexible curriculum. In this regard, the topic of life purpose becomes especially relevant given that it is with a mentor and through practical exercises regarding the subject that students determine their life purpose statement. With this statement, students make decisions regarding the future selection of courses that bring them closer to developing the professional skills needed to fulfil their purpose; this exercise is based on the purpose model described by Michael Steger (Steger et al., 2008). Since students start university, it has been proposed that individuals should make decisions based not on professional success but on the combination of personal strengths, what he or she is passionate about and what will generate value for his or her community.

(b) The principles of wellbeing and happiness course

The Principles of Wellbeing and Happiness course is the cornerstone of the Tecmilenio University curriculum model at the undergraduate level. This is a 16-week course that all students take in their first semester at the university and consists of the basic positive psychology topics using the PERMA model (Seligman, 2012), in addition to a detailed analysis focusing on strengths (Peterson & Seligman, 2004). This course is taken by the bulk of students in the fall given that the fall semester is when entry is greatest. During the COVID-19 pandemic in Mexico (beginning in March 2020 and still occurring at the time of writing this chapter), students at the undergraduate level had nearly completed the course.

The instructional design of the course includes an explanation of the topic and exercises that students must complete individually. That is, students seek to understand why the topics are relevant and always have practical exercises. The evaluations during the course occur through reflection. In the design of this course, it was recognized that it is not possible to evaluate whether an exercise such as a gratitude journal or a portfolio of positive emotions is well done; therefore, each student's reflection on their own understanding of wellbeing and how to generate it is evaluated.

These are the topics that make up the course: introduction to the scientific study of happiness, the components of wellbeing (PERMA), positive emotions, optimism, pleasure and joy for life, character strengths, optimal experiences (flow), positive relationships, gratitude, purpose and meaning of life, resilience, goals and achievements, perseverance and self-regulation, mindfulness, social wellbeing and life purpose component during community service.

Mexican legislation requires 600 hours of community service to obtain a bachelor's degree. Tecmilenio University made the decision that this service should be oriented towards teaching vulnerable populations (primary and secondary students of public schools in low-income areas) the topic of wellbeing, selecting the programme Aulas Felices (Arguis et al., 2010). Additionally, this study shows how helping others can bring wellbeing to their own lives. To date, 4,200 students per year participate in community service.

Another of the community service programmes that are related to positive psychology is positive museums. In association with a cultural festival in the city, students guided by staff of the Institute for Wellbeing and Happiness generate tours and activities in which museum visitors can explore their positive emotions through art, history and science. Currently, there are 11 museums in the city of Monterrey and approximately 15,000 visitors annually participate in positive museum activities.

During the COVID-19 pandemic, community service was interrupted because it was not possible to attend schools, museums and other organizations in person. During this period, students voluntarily taught lessons about the programme to children of university collaborators, with an average participation of 400 children.

(c) Business semester and the positive organizations course

All university students, according to the university plan, must complete a practical internship in a company for six months; the purpose is two-fold: developing professional skills and incorporating positive elements into the company. During the pandemic in Mexico, 2,073 students were enrolled in the business semester; these students had to follow the rules of the industries and states where they were located.

Generally, during the internship, students are supported and monitored by a business coach, who is a professor who guides their learning process in their professional training. They are taking the Positive Organizations course, which is based on the Healthy and Resilient Organization model (Salanova, 2008). In this course, students must first learn the Healthy and Resilient Organization model and then perform an intervention, in which they have the option of choosing between three pathways for their work team. These pathways focus on strengths (Peterson & Seligman, 2004), positive relationships at work (Seligman, 2012) and meaningful work (Dik et al., 2013). This course is always taken in an online format because it is a complement to the business semester, during which students do not visit the campus or the headquarters at which he/she is enrolled during that period. For the development of this course, the idea is that students will learn and deepen their development strategy skills by teaching others about topics related to positive organizations.

(d) Student wellbeing and development activities

During the course of their studies, university students have access to sports, cultural programmes and student clubs. At the high school level, a redesign of student activities was carried out to create a system of houses or fraternities similar to the English model. With this model, it is intended that students generate a network of positive relationships through student activities. Students have a sense of belonging to their group, and within cultural, sports and academic clubs, processes of competition and coexistence are generated. Students also undertake a community support project during high school. Students in their final year are the captains of the societies and transfer this leadership to their classmates each year.

There is an intensive robotics programme with regional, national and international tournaments in which students participate. In these leagues, university students have taken the Wellbeing and Happiness Ecosystem to other spaces and shared it with universities throughout Mexico and the world.

At the university level, there are clubs by academic area, the discipline of study and interest. In addition, some student leadership events, such as the "Empower" congress, promote participation in community entrepreneurship projects. Both at the undergraduate and high school levels, students from all campuses of the university are invited to national events, including national sports and cultural events, a student leadership event and an event on wellbeing and happiness organized by the Institute for Wellbeing and Happiness.

Currently, the university has students in 31 cities. There are two annual academic cycles. The first is for young people from 15 to 23 years old; the cycle runs from August to December and from January to May of each year. In this calendar cycle, there are high school students and undergraduate level students. The second calendar cycle is for adults or executives and runs from September to December, from January to April and from May to August. In-person activities are primarily designed for young students, with exercises, networking meetings and mindfulness practices incorporated into classrooms for adult students.

During the pandemic, sports and cultural classes were converted into an online model, with daily offerings by the university's social networks and in closed classes. As an additional measure, an 8-week mindfulness course based on character strengths was made available to the entire university and translated for the university with the authorization of the VIA Institute (Ivtzan et al., 2016).

Importantly, professors and collaborators are also part of the Wellbeing and Happiness Ecosystem of the university. All collaborators at the university take the Foundations of Positive Psychology course. From 2013 to date, 7,054 university collaborators, either professors or administrators, have completed the 12-week programme. In the period from 2018 to 2019, scholarships for this programme were also extended to the relatives of the collaborators to generate a community based on positive psychology. During the period of confinement, programmes have been deployed to allow working from home, and these have been accompanied by additional training, such as an 8-week course on mindfulness and topics related to work flexibility. In addition, activities were designed to support collaborators who are home-schooling their children.

A wellness programme, with small wellness interventions at the beginning of class and/or at closing, was implemented very quickly for all courses taught remotely. Twelve short mindfulness audio exercises, one-minute videos with positive emotion exercises, stretching and strategies to generate positive relationships were made available so that students and professors continued to cultivate the Ecosystem from home.

From March to August 2020, 40 webinars were also given with the aim of educating the entire community regarding issues related to positive organizations, mindfulness, emotional education, positive families, sleep management and healthy stress management during the period of confinement. This was designed as a community outreach programme, and it was hoped that collaborating students, professors, parents of the students and alumni would participate.

There was also another community activity designed by the university that occurred during the pandemic. In the city of Monterrey, where our main campus is located, in a previously made agreement with city museums that has continued throughout the pandemic, virtual tours for the community based on positive emotions were made available, with the goal of combining art and wellbeing through a virtual intervention. To date, 5,000 people have viewed the positive museum tours.

The effect of the Tecmilenio University Wellbeing and Happiness Ecosystem, which incorporates life purpose as its central axis, is also reflected in its graduates. A survey conducted in 2020 identified that university graduates who developed a better-defined life purpose experience a series of heliotropic effects at the personal, family and work levels. Cameron (2008) likens the "heliotropic effect" as being the natural tendency of humans to find elements that enrich our lives and lead us to live them better, drawing closer to positive energies and away from negative energies. The main findings of the study suggest that graduates of Tecmilenio University who have a more defined purpose manifest multiple positive side effects (heliotropic), in terms of having lower chances of being unemployed or recently losing employment as well as greater chances of finding themselves in a higher income bracket. Notably, having a better-defined purpose also impacts the personal level because it is positively correlated with greater satisfaction with life and good health and having more friends and family willing to provide support in emergency situations (Charles-Leija et al., 2020).

The present study

This chapter is necessary because it is worth investigating the wellbeing of young people in a country such as Mexico, a nation that has unique characteristics. The pandemic generated a health crisis in a country that was already experiencing severe problems in terms of economy and violence. In August 2020, Mexico was among the three countries with the highest number of deaths associated with COVID-19 worldwide. In this context, it is essential for universities to have wellbeing and happiness ecosystems that can be applied in virtual spaces and contribute to students being able to better face the difficulties associated with the pandemic. This study seeks to answer three research questions: What is the impact of COVID-19 on the wellbeing of students at Tecmilenio University? What is the effect of COVID-19 on the positive emotions of students? Is there a difference in academic levels?

Methods and data sources

This chapter presents data obtained in the spring session, during the month of April for 2018, 2019 and 2020. The most recent survey was carried out amid the COVID-19 pandemic. Although Tecmilenio University has a population obtaining master's and specialty degrees, the high school and professional student populations have more involvement with the wellbeing and happiness ecosystem developed by the institution. This is why the present analysis focuses on such educational levels.

Table 3.2 presents the number of students who participated by answering the wellbeing survey. Student participation was highest for 2019. However, representative samples were obtained for all periods. The instrument used to capture the wellbeing of students is the PERMA Profiler, developed by Butler and Kern (2016). It consists of 23 items that measure seven dimensions, the five corresponding to the PERMA proposed by Seligman (2012, 2018), negative emotions (N) and health (H), as well as an item for the overall happiness (hap). For each element of PERMA, there are three items; for health, there are three items; and for negative emotions, there are four items. Next, the PERMA Profiler questionnaire is described (Table 3.3). The questions are answered using a scale that ranges from 0 to 10.

Table 3.2 Students who responded to the survey, spring 2018, 2019 and 2020

Period	*High School*	*Professional*
2018	2,853	1,226
2019	4,949	2,354
2020	1,345	914

Source: Opin@ survey

Table 3.3 Description of the PERMA Profiler questionnaire items

Original Question	*Name of the Variable*
In general, how often do you feel joyful?	P1
In general, how often do you feel positive?	P2
In general, to what extent do you feel content?	P3
How often do you become absorbed in what you are doing?	E1
In general, to what extent do you feel excited and interested in things?	E2
How often do you lose track of time while doing something you enjoy?	E3
To what extent do you receive help and support from others when you need it?	R1
To what extent do you feel loved?	R2
How satisfied are you with your personal relationships?	R3
In general, to what extent do you lead a purposeful and meaningful life?	M1
In general, to what extent do you feel that what you do in your life is valuable and worthwhile?	M2
To what extent do you generally feel you have a sense of direction in your life?	M3
How much of the time do you feel you are making progress towards accomplishing your goals?	A1
How often do you achieve the important goals you have set for yourself?	A2
How often are you able to handle your responsibilities?	A3
In general, how would you say your health is?	H1
How satisfied are you with your current physical health?	H2
Compared to others of your same age and sex, how is your health?	H3
In general, how often do you feel anxious?	N1
How lonely do you feel in your daily life?	N2
In general, how often do you feel angry?	N3
In general, how often do you feel sad?	N4
Taking all things together, how happy would you say you are?	Hap

Source: Opin@ survey

Figure 3.2 shows the correlations among the 23 items of the PERMA Profiler, showing two large groupings: 19 items corresponding to elements of student wellbeing, both high school and professional, and a correlation among elements of discomfort. From the illustration, it can be observed that the overall happiness variable presents a strong correlation with positive emotion variables (feeling joyful, content and positive) and, in a second level, with E2 (feeling excited and interested in things), R3 (feeling satisfied with personal relationships) and A2 (achieving goals). All correlations are significant at 0.01.

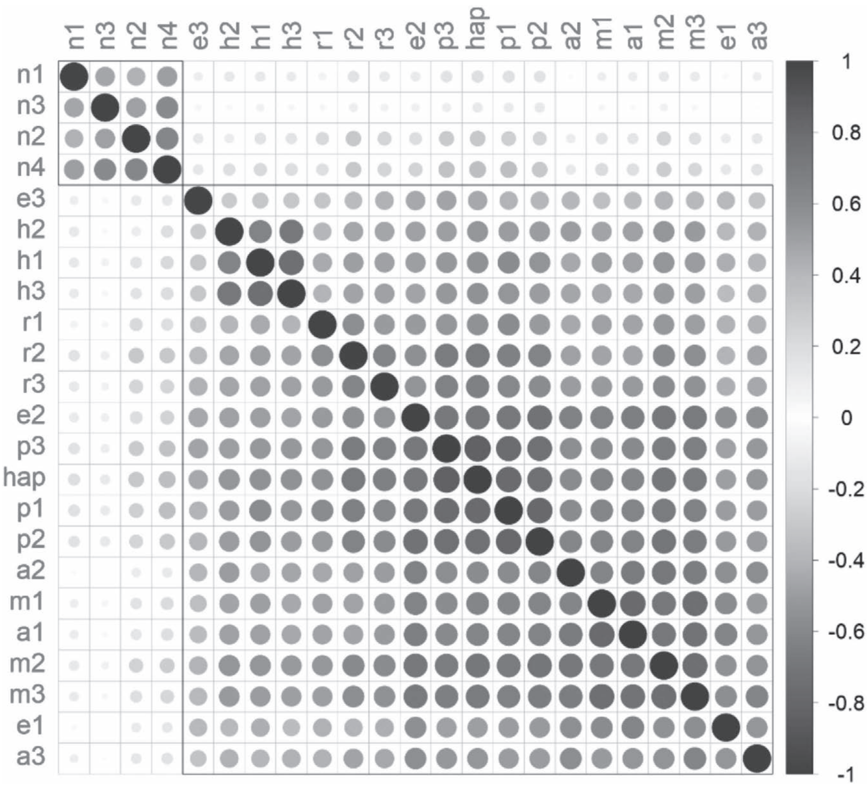

Figure 3.2 Correlations among the 23 items of the PERMA Profiler, 2018–2020

Source: Opin@ survey

Figure 3.3 shows the correlations in four groups. Negative emotions represent one group; E3 (losing track of time when doing an enjoyable activity) represents a group by itself; health represents a group by itself; and the remaining items, being very similar to each other, form a grouping representing the overall wellbeing. The illustration corresponds largely to the theoretical postulates of PERMA: although health is a fundamental element of the overall wellbeing of individuals, the five blocks that correspond to positivity, engagement, relationships, meaning and achievement are elements more integrated within themselves as pillars of wellbeing (Seligman, 2018). Item E3 is not fully integrated into the overall block of wellbeing items, perhaps because this flow component is not as close to the other items for the overall happiness.

Results

Tables 3.4 and 3.5 present the averages for the PERMA components for high school and professional students. For ease of interpretation, only the years 2019

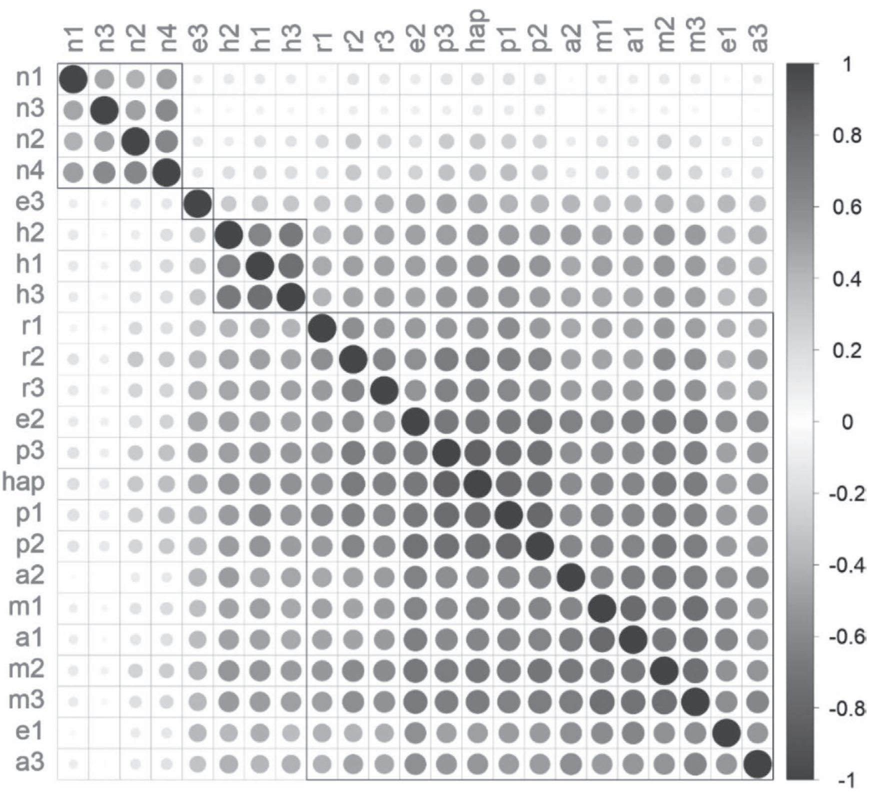

Figure 3.3 Correlations among the 23 items of the PERMA Profiler, four groups, 2018–2020

Source: Opin@ survey

and 2020 are contrasted. The majority of the 15 items show a negative effect caused by the pandemic. The only PERMA domains for which adverse effects are not shown is in the third item, engagement, which corresponds to the perception that time is passing very quickly when engaged in an activity, as well as in the last item, achievement, corresponding to feeling able to handle responsibilities. For high school students, the first question of the domain is "meaning", specified as follows: "In general, to what extent do you lead a purposeful and meaningful life?". This is a highly relevant result for Tecmilenio University because "purpose" is central to its Wellbeing and Happiness Ecosystem, and this suggests that high school students do not perceive that their life purpose has been impacted by the pandemic. Several studies have shown the importance of adolescents having a well-defined life purpose (Brassai et al., 2011; Damon et al., 2003).

Regarding the other elements measured by the PERMA Profiler, that is, health, negative emotions and happiness, the results showed interesting improvements.

Table 3.4 Averages for PERMA elements by academic level, 2019 and 2020

		High School			*Professional*		
	Period	*Mean*	*Std. Dev.*	*Sig*	*Mean*	*Std. Dev.*	*Sig*
Joy	2019	8.32	0.02	***	8.42	0.03	***
	2020	8.15	0.05		8.20	0.06	
Positive	2019	8.19	0.03	***	8.43	0.04	***
	2020	8.03	0.05		8.25	0.06	
Content	2019	8.50	0.02	***	8.59	0.03	***
	2020	8.33	0.05		8.39	0.06	
Absorbed	2019	8.28	0.02	**	8.48	0.03	***
	2020	8.19	0.05		8.19	0.06	
Excited	2019	8.41	0.02	***	8.58	0.03	***
	2020	8.26	0.05		8.42	0.06	
Time_passes_q	2019	8.99	0.02		8.93	0.03	
	2020	9.00	0.04		8.95	0.05	
Help_from_others	2019	8.31	0.03	***	8.46	0.04	***
	2020	8.11	0.06		8.22	0.07	
Feel_loved	2019	8.52	0.03	**	8.65	0.04	**
	2020	8.44	0.05		8.50	0.06	
Satisfied_relat	2019	8.48	0.03	**	8.51	0.04	***
	2020	8.35	0.05		8.31	0.07	
Purpose_meaning	2019	8.19	0.03		8.59	0.03	***
	2020	8.12	0.06		8.35	0.06	
Life_worth_living	2019	8.29	0.03	**	8.61	0.04	***
	2020	8.16	0.06		8.39	0.06	
Life_direction	2019	8.17	0.03	***	8.45	0.04	***
	2020	7.96	0.06		8.25	0.06	
Achievement_goals	2019	8.03	0.03	***	8.32	0.04	***
	2020	7.88	0.06		8.02	0.06	
Frequency_goals	2019	8.08	0.03	*	8.28	0.03	***
	2020	8.00	0.05		8.02	0.06	
Responsibilities	2019	8.41	0.02		8.61	0.03	
	2020	8.37	0.05		8.55	0.05	
Sample	2019	4,949			2,354		
	2020	1,345			914		

Source: Own elaboration with data obtained from the four-month student survey. Significance levels: *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

High school students reported better-perceived health than other people of the same age. It is striking that for both high school and professional students, three of the four negative emotions showed a statistically significant reduction. The students showed lower levels of anxiety, loneliness and anger in April 2020 compared to the same period of the previous year.

A methodological challenge presented by the study is that Likert-type measurements of wellbeing regularly present bias towards high values, that is, eight, nine and ten; therefore, it is difficult to observe significant increases when scores are already high. In 2019, all the values for the wellbeing variables were higher than eight. As given earlier, the increase seen in the health scores and the reduction in the negative emotions is all the more remarkable.

Table 3.5 Averages for health, negative, happiness elements by academic level, 2019 and 2020

		High School			*Professional*		
	Period	*Mean*	*Std. Dev.*	*Sig*	*Mean*	*Std. Dev.*	*Sig*
Health	2019	8.44	0.02	***	8.40	0.04	
	2020	8.58	0.05		8.43	0.06	
Health-satisfaction	2019	7.65	0.03	*	7.60	0.05	**
	2020	7.53	0.07		7.43	0.08	
Comparative_health	2019	8.29	0.03	*	8.27	0.04	
	2020	8.37	0.05		8.25	0.07	
Anxiety	2019	7.02	0.04	***	6.80	0.06	***
	2020	6.82	0.07		6.46	0.09	
Loneliness	2019	5.35	0.05	*	5.22	0.07	**
	2020	5.19	0.09		4.93	0.11	
Anger	2019	5.40	0.04	**	5.36	0.06	***
	2020	5.23	0.08		5.00	0.09	
Sadness	2019	5.16	0.04		4.97	0.06	
	2020	5.09	0.08		4.87	0.10	
Happiness	2019	8.43	0.03	***	8.56	0.03	***
	2020	8.24	0.05		8.38	0.06	
Sample	2019	4,949			2,354		
	2020	1,345			914		

Source: Own elaboration with data obtained from the four-month student survey. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

The results found are consistent with expectations. On the one hand, previous studies of wellbeing have found that it remains stable over time. Due to the emergency conditions generated by COVID-19, the average person suffers various types of reductions in their level of wellbeing. An unexpected element was observed lower levels of negative emotions with respect to 2019. This result may be attributed to students implementing the emotional tools that allow them to better cope with crisis situations. Some elements of wellbeing, scored high in 2019, were lower in 2020, and discomfort and negative emotions did not increase; in fact, they decreased.

This chapter was motivated by three research questions: What is the impact of COVID-19 on the wellbeing of the students at Tecmilenio University? What is the effect of COVID-19 on the positive emotions of the students? Is there a difference between academic levels?

Significance of the study

Regarding the first and second questions, the results show that the overall health emergency affected the wellbeing of students. Most of the items associated with the PERMA elements are lower in 2020 than in 2019, as would be expected. A notable finding, however, is the reduction in negative emotions. This result can be linked to the university's response to its students, offering the possibility of continuing most of their school activities virtually. The university also provided students, through care and consulting services, access to mental health resources as well as prevention tips for coping with the health emergency.

Regarding the third research question, the third item associated with health stands out. In this item, students are questioned about their health in relation to individuals of the same age; high school students showed an improvement in 2020 compared to the previous year. This result can be attributed to the fact that in April, the youngest students at the university noticed that their peers from other universities had poorer health. This variable was not significant for professional-level students.

The analyses performed for the two academic levels suggest that the effects of the COVID-19 pandemic, specifically health issues, are greater in high school students. When asked "In general, how is your health?" the high school students gained a better perception of their health over the period considered. The impact of the remaining variables was very similar for both populations.

Conclusions

The purpose of this study was to explore the level of wellbeing of students, during the COVID-19 emergency, at a university focused on positivity. The data allowed exploring changes in the levels of wellbeing of professional and undergraduate students and finding positive effects from having been immersed in a wellbeing and happiness ecosystem.

A significant contribution of this study is that it provides confirmation that it is important to establish a measure of the wellbeing of students across time. With this, it is possible to gauge their emotions and take actions to improve their quality of life. Another relevant contribution of the present study is that it showed various benefits from using the PERMA Profiler instrument in a university context. The instrument allowed groupings based on the constructs "wellbeing" (the elements that make up the PERMA and the health component) and "discomfort" (the four negative emotions evaluated). Furthermore, the instrument allowed precisely distinguishing that health and flow have characteristics of wellbeing that are slightly different from the main blocks of wellbeing (Seligman, 2018).

The results previously described highlight that the Wellbeing and Happiness ecosystem at Tecmilenio University is fulfilling its mission of providing students with tools to face various personal and professional challenges. The COVID-19 health emergency has been a challenge for people around the world in terms of mental and emotional health. The students at Tecmilenio University appear to have the wellbeing skills to adequately cope with negative emotions. Emotions such as anxiety, loneliness and anger were lower in April 2020 than in April 2019, and sadness did not change significantly.

In terms of public policy, it is crucial that governments consider incorporating elements of positive education in educational programmes at all levels. Tecmilenio University focuses its efforts on high school, professional and master's students. However, positive education can positively impact early ages as well.

The case of Mexico is a topic of great interest because it is a developing country with many social problems to address. Bullying in Mexico is severe and can be the trigger for criminal activity in the country. Positive education can serve as a fundamental tool to address this problem. Training students from a foundation of empathy and teaching them to recognize their own strengths and those of others can foster better coexistence at school at all educational levels, especially in middle and high school, which represent key training stages for young people.

The general population should integrate themselves into strength-based mindfulness programmes. The efforts made at Tecmilenio University in this area show that practices focusing on strengths not only reduce stress (a goal that other workshops also achieve) but also contribute to improving the wellbeing of those who carry out the practices. Future research could implement different instruments to measure the wellbeing of students.

Ethics: The Tecmilenio University's Office of Research Ethics, Compliance and Integrity approved this study (Approval No: 2016–001).

Acknowledgements: The authors thank Mario Toledo for his support in this chapter elaboration. This research was funded by PROEZA.

References

Arguis, R., Bolsas, A. P., Hernandez, S., & Salvador, M. (2010). *Programa aulas felices*. SATI.

Brassai, L., Piko, B. F., & Steger, M. F. (2011). Meaning in life: Is it a protective factor for adolescents' psychological health? *International Journal of Behavioral Medicine*, *18*(1), 44–51. https://doi.org/10.1007/s12529-010-9089-6

Butler, J., & Kern, M. L. (2016). The PERMA-Profiler: A brief multidimensional measure of flourishing. *International Journal of Wellbeing*, *6*(3), 1–48. https://doi.org/10.5502/ijw.v6i3.526

Cameron, K. (2008). *Positive leadership*. Berrett-Koehler.

Charles-Leija, H., Aboites, G., & Llamas, I. (2018). Una revisión de aportaciones que contribuyeron al estudio de la utilidad y la felicidad en la economía. *Análisis Económico*, *XXXIII*(84), 57–76.

Charles-Leija, H., Mora, C., & Toledo, M. (2020). Positive side effects of having a life purpose, the case of graduates of Tecmilenio University. *Congreso Internacional de Innovación Educativa, 7*, 1323–1328.

Damon, W., Menon, J., & Cotton Bronk, K. (2003). The development of purpose during adolescence. *Applied Developmental Science*, *7*(3), 119–128.

Dik, B., Byme, Z. S., & Steger, M. (2013). *Purpose and meaning in the workplace*. American Psychological Association.

Duckworth, A. L., Grant, H., Loew, B., Oettingen, G., & Gollwitzer, P. M. (2011). Self-regulation strategies improve self-discipline in adolescents: Benefits of mental contrasting and implementation intentions. *Educational Psychology*, *31*(1), 17–26.

Gable, S. L., Reis, H. T., Impett, E., & Asher, E. (2004). What do you do when things go right? The intrapersonal and interpersonal benefits of sharing positive events. *Journal of Personality and Social Psychology*, *87*(2), 228.

Goleman, D., & Davidson, R. (2017). *The science of meditation: How to change your brain, mind and body*. Penguin.

Ivtzan, I., Niemiec, R. M., & Briscoe, C. (2016). A study investigating the effects of Mindfulness-Based Strengths Practice (MBSP) on wellbeing. *International Journal of Wellbeing*, *6*(2).

Layard, R. (2020). *Can we be happier?* Penguin.

Oades, L., Robinson, P., Green, S., & Spence, G. (2011). Towards a positive university. *The Journal of Positive Psychology*, *6*(6), 432–439.

Peterson, C. (2006). *A primer in positive psychology*. Oxford University Press.

Peterson, C., & Seligman, M. (2004). *Character strengths and virtues: A handbook and classification*. Oxford University Press.

Rojas, M., & Charles-Leija, H. (2021). Chile, milagro de crecimiento económico, pero . . . ¿y el bienestar? *Perfiles Latinoamericanos*.

Salanova, M. (2008). Organizaciones saludables, una aproximación desde la psicología positiva. En *Psicología Positiva Aplicada* (pp. 403–427). Alianza Editorial.

Seligman, M. (2012). *Flourishing: A visionary new understanding of happiness and well-being*. Simon & Schuster.

Seligman, M. (2018). PERMA and the building blocks of well-being. *Journal of Positive Psychology*, *13*(4), 333–335. https://doi.org/10.1080/17439760.2018.1437466

Seligman, M., & Adler, A. (2019). *Positive education*. Global Happiness Policy Report.

Sheldon, K. (2014). Becoming oneself: The central role of self-concordant goal selection. *Personality and Social Psychology Review*, *18*(4), 349–365.

Steger, M. F., Kashdan, T. B., Sullivan, B. A., & Lorentz, D. (2008). Understanding the search for meaning in life: Personality, cognitive style, and the dynamic between seeking and experiencing meaning. *Journal of Personality*, 199–228.

Steger, M. F., Oishi, S., & Kashdan, T. B. (2009). Meaning in life across the life span: Levels and correlates of meaning in life from emerging adulthood to older adulthood. *The Journal of Positive Psychology*, *4*(1), 43–52.

Stravos, J., Cooperrider, D., & Kelley, D. L. (2003). Strategic inquiry appreciative intent: Inspiration to SOAR, a new framework for strategic planning. *AI Practitioner*, (11).

Waters, L., Barsky, A., & McQuaid, M. (5/2 de 2012). Positive education: A whole-school approach. *International Positive Psychology Association Newsletter*, *5*(2), 1–2.

Wesson, K., & Boniwell, I. (2007). Flow theory – its application to coaching psychology. *International Coaching Psychology Review*, *2*(1), 33–43.